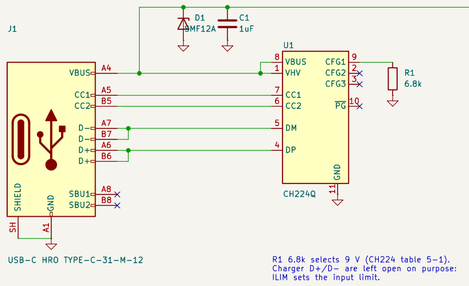
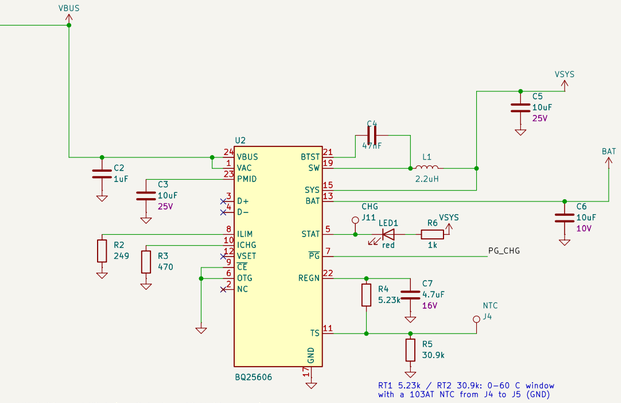
USB-C input, PD sink (requests 9 V)



R1 6.8k selects 9 V (CH224 table 5-1).
Charger D+/D- are left open on purpose:
ILIM sets the input limit.

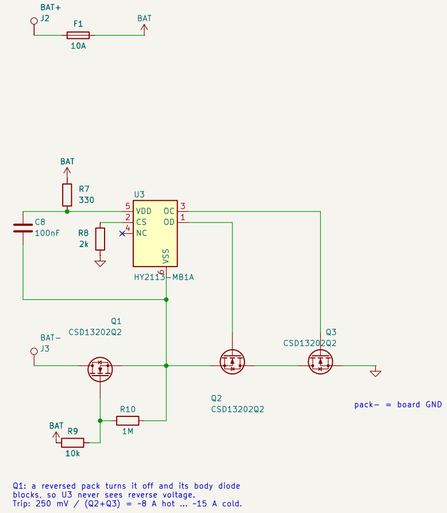
Charger BQ25606: standalone NVDC, 1.44 A charge, 1.92 A input limit



ILIM 249R -> 1.92 A (<= 2 A of an 18 W PD charger)
ICHG 470R -> 1.44 A VSET open -> 4.208 V

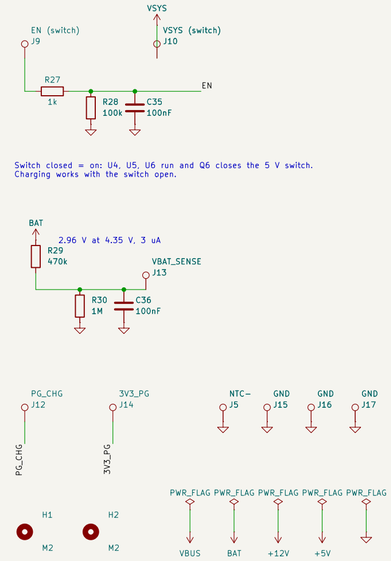
RT1 5.23k / RT2 30.9k: 0-60 C window
with a 103AT NTC from J4 to J5 (GND)

1S2P pack: fuse, reverse block, HY2113 protector

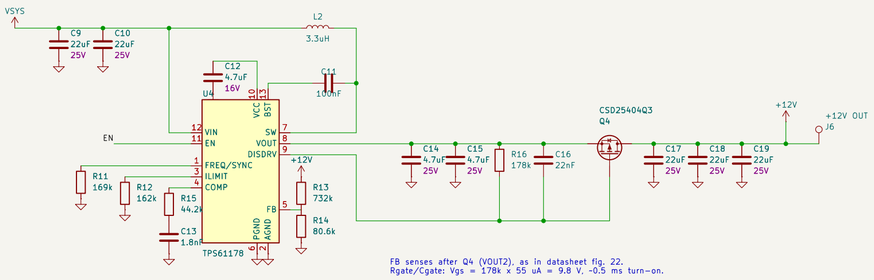


Q1: a reversed pack turns it off and its body diode
blocks, so U3 never sees reverse voltage.
Trip: 250 mV / (Q2+Q3) = ~8 A hot ... ~15 A cold.

Solder pads, enable switch, battery sense

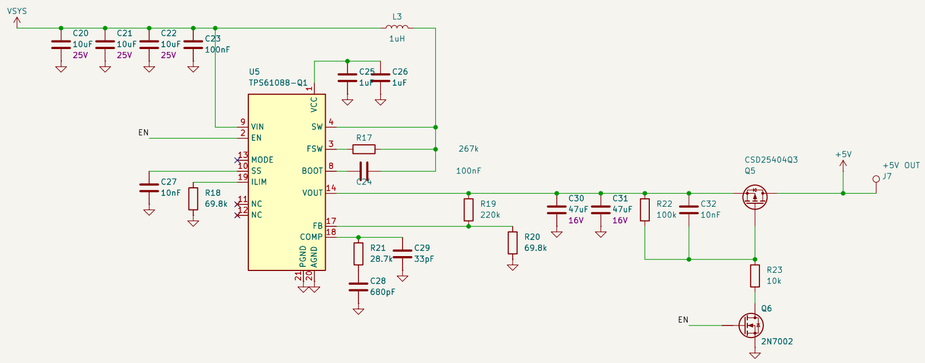


12 V / 0.5 A boost TPS61178 (WEBENCH design 33)



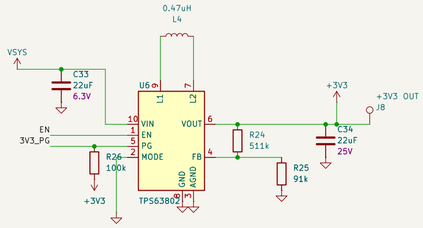
FB senses after Q4 (VOUT2), as in datasheet fig. 22.
Rgate/Cgate: Vgs = 178k x 55 uA = 9.8 V, ~0.5 ms turn-on.

5 V / 2.5 A boost TPS61088-Q1 (WEBENCH design 37) + load switch



ILIM = 550000 / 69.8k = 7.9 A (-Q1 has a 0.6 V ILIM reference)
FSW 267k (to SW) = 594 kHz, SS 10 nF = 2.4 ms

3.3 V / 1.5 A buck-boost TPS63802 (WEBENCH design 36)



Sheet: /	
File: minipc-power.kicad_sch	
Title: minipc-power: USB-C PD / 1S2P power board	
Size: A2	Date: 2026-10-24
KiCad E.D.A. 10.0.1	**Rev: 1**
	Id: 1/1